Problem Definition

Accurate segmentation of agricultural land from satellite imagery is a critical task in modern precision farming, resource monitoring, and environmental planning. Recent advances in remote sensing and machine learning, particularly deep learning-based semantic segmentation, have enabled automated analysis of large-scale agricultural regions. However, challenges such as spectral variability, seasonal changes, and mixed land cover types demand robust and adaptable models. Leveraging multi-spectral data from satellites like Sentinel-2, along with deep neural architectures, can provide meaningful insight for agricultural land management.

Literature Review

Several recent studies have investigated deep learning approaches for agricultural image segmentation. In [1], attention-based semantic segmentation was used for crop mapping with Sentinel-2 imagery. Study [2] presented a comparative analysis of segmentation architectures including U-Net, SegNet, DeepLab, and TransUNet, confirming the effectiveness of encoder–decoder baselines. Zhao et al. [3] demonstrated the applicability of U-Net on Sentinel-2 data using patch-based training with strong performance on Dice score. Two survey papers [4, 5] provide comprehensive reviews of existing techniques, datasets, and evaluation metrics, offering guidance on best practices for agricultural segmentation pipelines.

Proposed Idea

This project aims to implement a deep learning-based image segmentation model for agricultural land classification using Sentinel-2 satellite imagery. After selecting a region of interest (ROI) in Iran, appropriate preprocessing techniques and vegetation indices will be applied. A neural network architecture will be designed or adapted based on insights from current research. The U-Net model will be trained using multi-spectral inputs to accurately identify classes such as vegetation, water bodies, and soil. Evaluation will be conducted using standard performance metrics including pixel accuracy, F1-score, Intersection over Union (IoU), and Kappa coefficient to ensure the reliability and practicality of the approach.

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[1] Gao, M., Lu, T., Wang, L. "Crop Mapping Based on Sentinel 2 Images Using Semantic Segmentation Model of Attention Mechanism," *Sensors*, 2023.

[2] Qi, L., et al. "Convolutional Neural Network Based Method for Agriculture Plot Segmentation in Remote Sensing Images," *Remote Sensing*, 2024.

[3] Zhao, K., et al. "Precision Agriculture: Crop Mapping using Machine Learning and Sentinel 2 Satellite Imagery," *arXiv*, 2023.

[4] Lei, L., et al. "Deep learning implementation of image segmentation in agricultural applications: a comprehensive review," *Artificial Intelligence Review*, 2024.

[5] Luo, Z., et al. "Semantic segmentation of agricultural images: A survey," *Information Processing in Agriculture*, 2024.